

4 Domestic goats, *Capra hircus*, show a wide range of coat patterns and colours. One gene involved in coat colour and pattern has multiple alleles. Four of these alleles are:

- **A**, the allele for white, is dominant to all others
- **A^b**, the allele for badgerface (stripes on face) and **A^g**, the allele for grey, are codominant
- **a**, the allele for black, is recessive to all others.

(a) State **all** the possible genotypes of:

a grey badgerface goat

a white goat.

[2]

(b) A cross between a black goat and a white goat produced a white goat. This white offspring was crossed with a grey goat. The genotype of the grey goat was not known.

Use genetic diagrams to show **all** the possible offspring genotypes and phenotypes that could result from the cross between the white offspring and the grey goat.

[4]

(c) When goat skin cells produce black melanin pigment these events occur:

- Melanocyte stimulating hormone (MSH) is secreted into the bloodstream.
- MSH binds to the melanocortin 1 receptor on the cell surface membranes of skin cells.
- This causes a rise in cyclic AMP within the cell.
- The enzyme tyrosinase is activated and begins melanin production.

The white allele **A** codes for a protein that binds irreversibly to the melanocortin 1 receptor. No rise in cyclic AMP occurs, even when MSH is present in the blood.

Explain the appearance of goats with genotype **aa**.

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